



Predicting Buckling Behavior in Naturally Curved Elastic Rods

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Background

Naturally curved elastic rods are 3D printed spirals with a parameterized radius, height, thickness, and length (See Figure 1). The radius also influences its “natural curvature,” which is defined as the inverse of its radius. When these rods are compressed, many experience snapping, or sudden buckling movements (See Figure 2), during both the loading and unloading phases [1]. The geometry of these elastic rods also affects their individual snap-through instabilities.

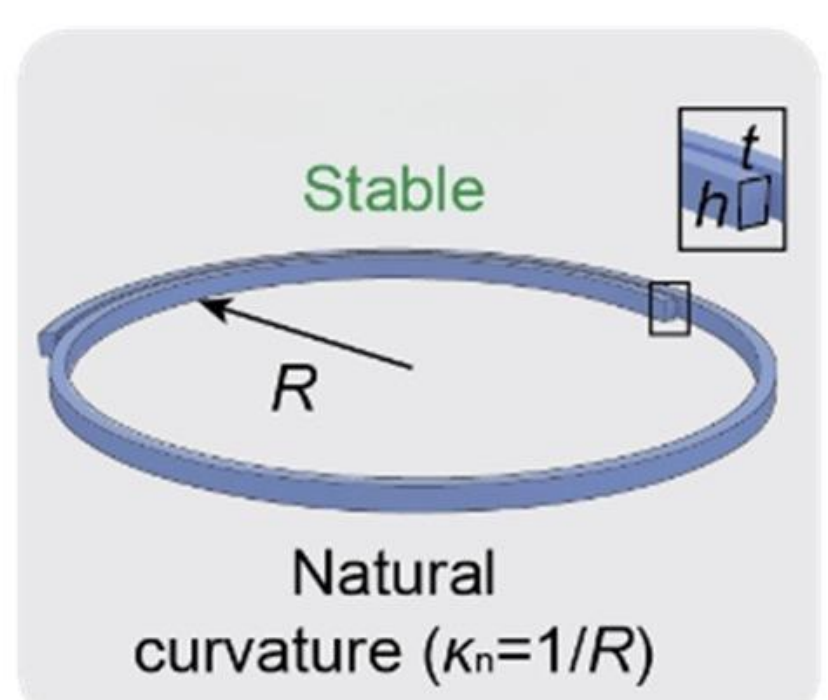


Figure 1: Naturally curved elastic rod

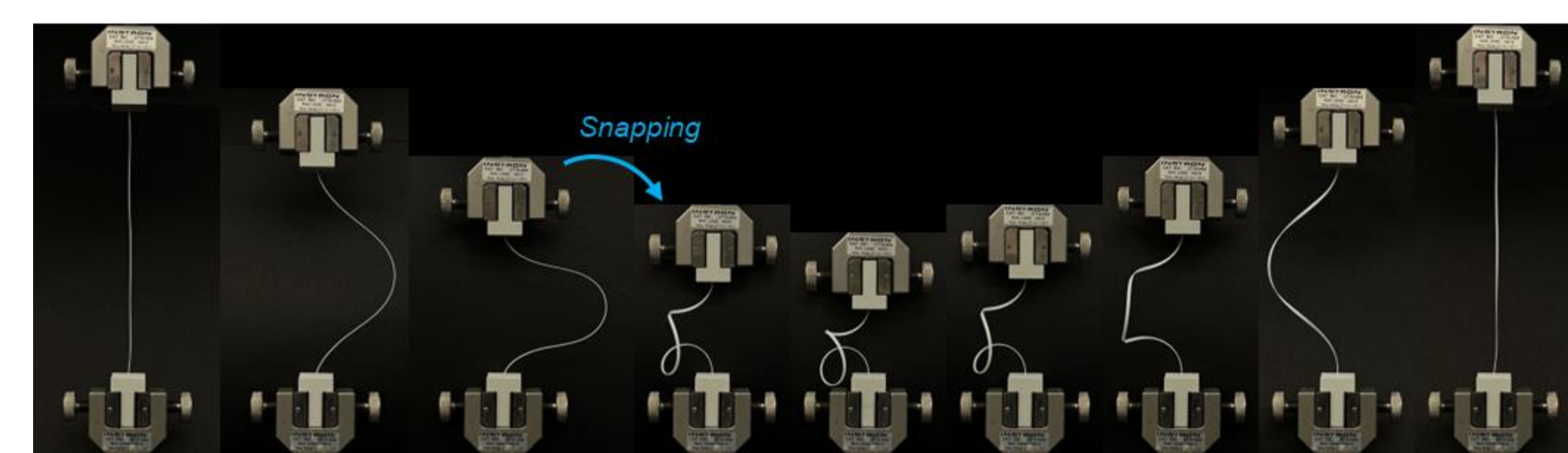


Figure 2: Case 2 Buckling with Snapping

Objective

The objective of this research is to be able to understand and predict a single given rod’s buckling behavior due to length, radius, height, and width. This understanding of rods at a component level will help inform potential system designs and applications.

Mechanical Testing Method

In general, little is known about naturally curved elastic rods at the quantitative level. To start informing a general understanding of rods, 50 semi-random samples with different radii, length, height,

and thickness were tested. For this first phase, 3D printing the rods with PLA has sufficient elasticity for the objective with an added ease of fabrication. To gather data, the rods are placed

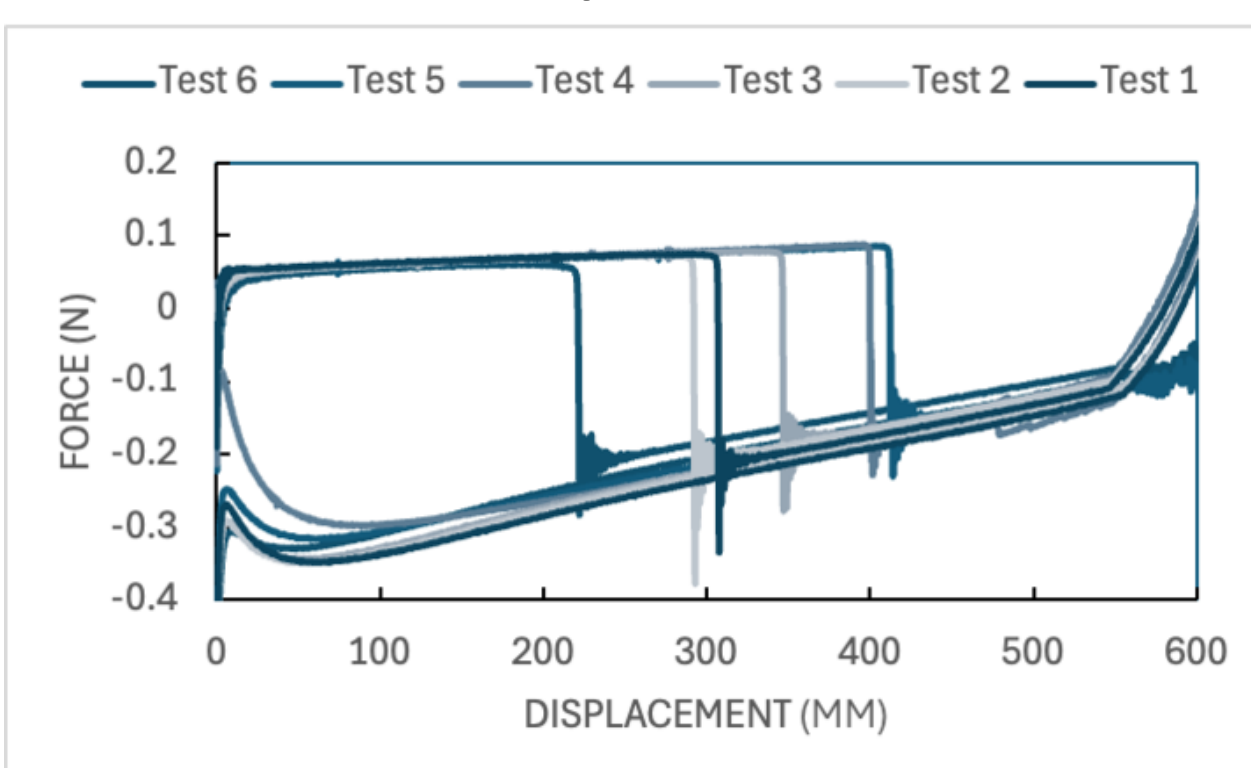


Figure 3: Force vs Displacement Graph

in an Instron Tensile Tester that cyclically compresses them six times to create a force vs displacement graph of all six tests, videoing the first and sixth tests. The graph generated (See Figure 3) by the cyclic compression test helps identify key buckling behaviors of the rod, like if it plastically deforms during the six tests, and whether it exhibits snapping behavior.

Sampling Method

Fully understanding the effect of geometric parameters on buckling behavior requires identifying the edge cases, or where a rod would change from buckling in one case to another. As more samples are added to the database, it will be easier to classify potential samples as certain cases before testing them. To get the most secure and accurate prediction, it is important to find the exact regions that determine a rod that will snap through, or not.

Buckling Case Classification

Individual unit rods buckle in primarily 4 different cases (See Figure 4).

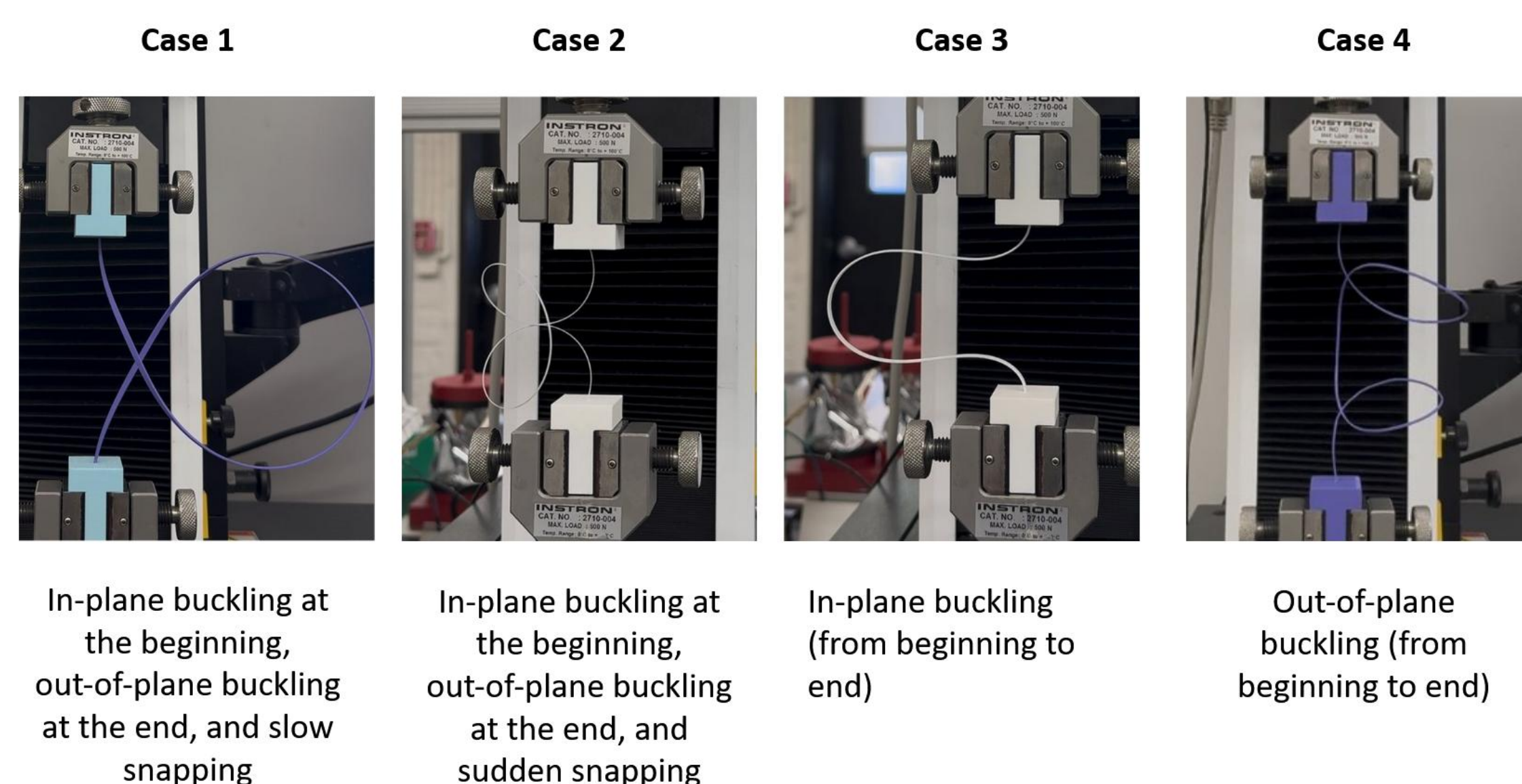


Figure 4: Four primary buckling cases of naturally curved elastic rods.

Of the 50 initial samples, 27 of them were classified as Case 3, 10 of them were classified as Case 4, and only 14 had any sort of snapping behavior. To achieve the goal of being able to predict how a given rod will buckle based on its geometric parameters, it is important to have an equal number of samples for each case. For the second phase of this investigation, intentionally targeting samples that will most likely snap will help increase the number of samples for Cases 1, 2, and 4. In fact, when lengthened, samples that were initially Case 3, or buckled in plane for both loading and unloading phases, would exhibit snapping behavior (See Figure 5).

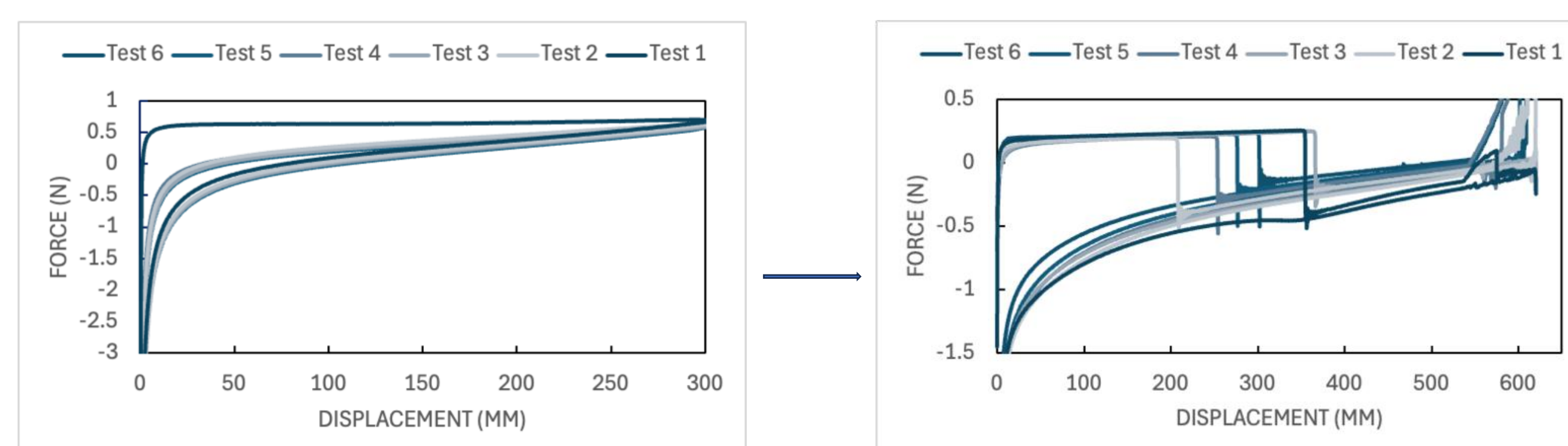


Figure 5: When lengthened, this rod changed from Case 3 to Case 2.

Extraneous cases, like those with a large length and small radius tend to buckle in unusual patterns with many loops. Further studies will be done to identify and define these cases. In addition, some samples exhibit snapping behavior during the unloading stage as well as the loading stage (See Figure 6). The rod would buckle around itself and uncoil with a snap, causing a change in the graph.

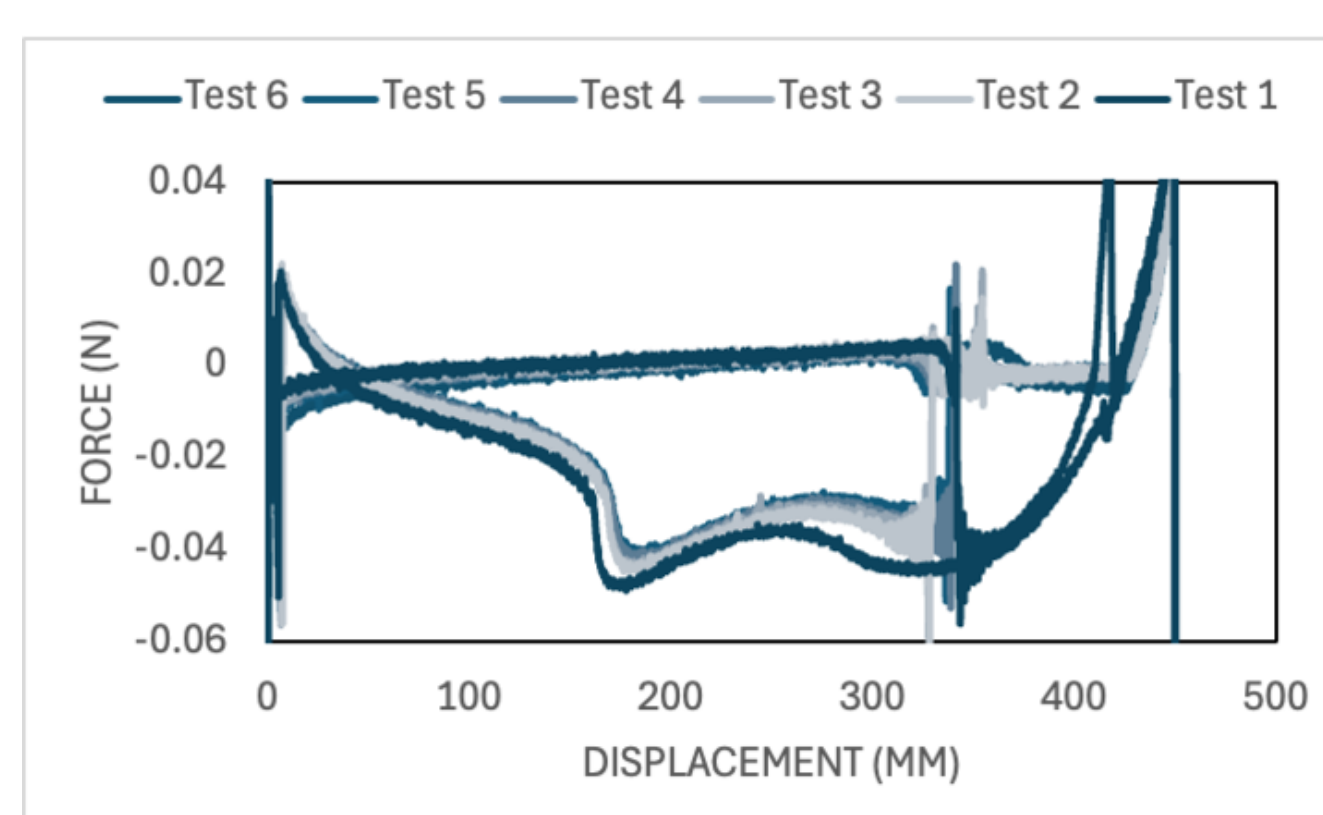


Figure 6: This sample exhibited snapping during both loading and unloading phases of the test



Figure 7: One sample with an unusual buckling formation

System Design

The understanding of rods and their stable states informs the creation and use of system designs such as Figures 8 and 9.

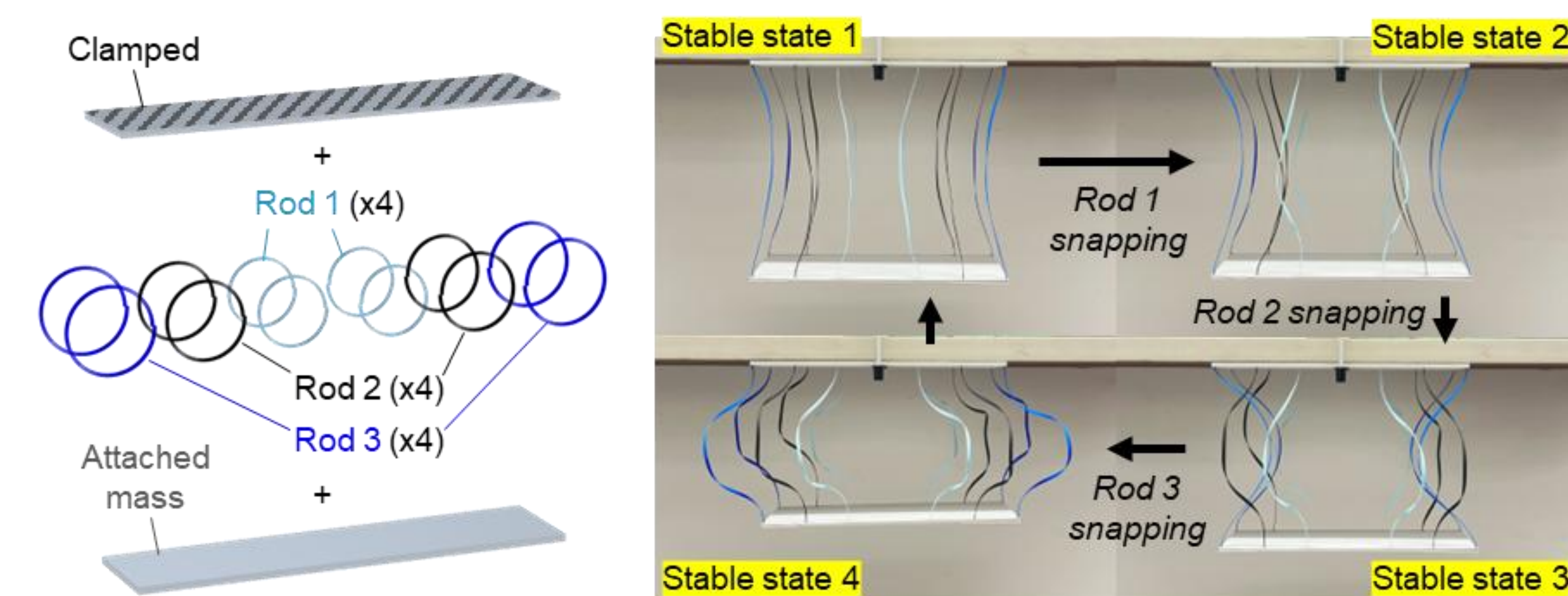


Figure 8: A 12-rod system of varying rod samples with four stable states

These systems have multiple stable states and can change shape because of the rods. These systems have varying oscillating behaviors due to their stable states.

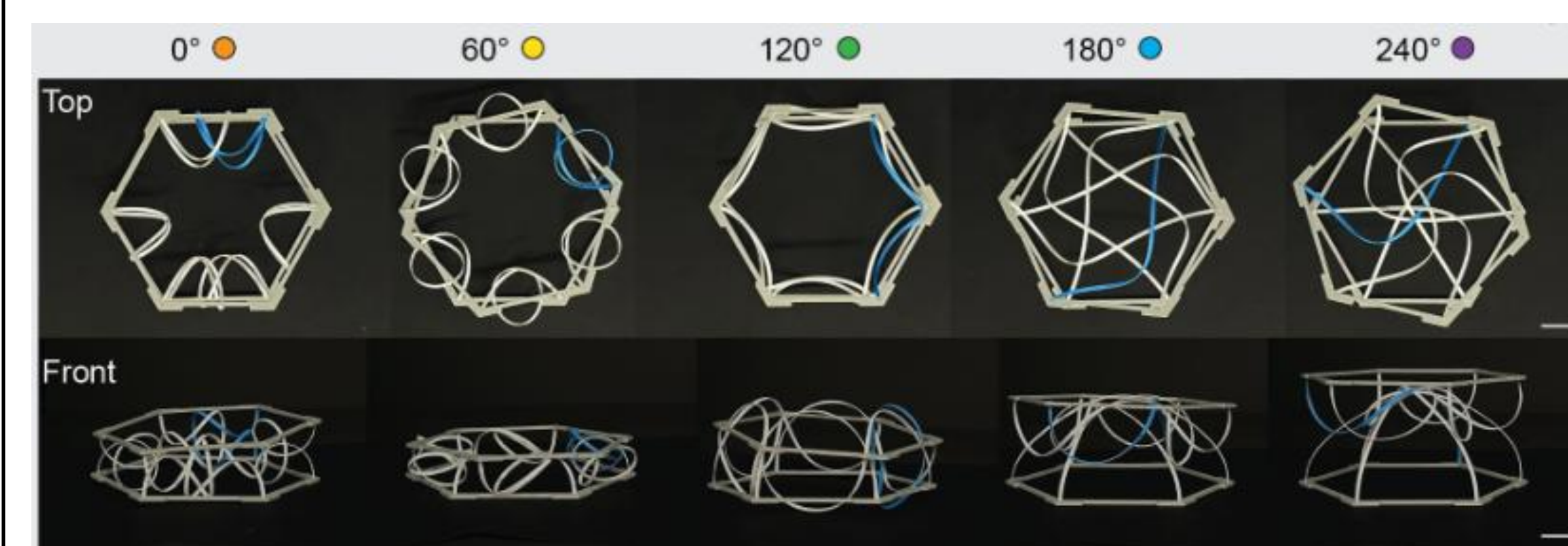


Figure 9: Hexagonal system with six identical rods and five stable states

Future Studies

For future studies, a method of removing more uncertainty during tests should be developed. In many cases, samples respond to tenths of Newtons of force disturbance, throwing off results. Rods are also useful in reconfigurable structures and robotics. Figure 10 shows a reconfigurable spinning system



that uses rods to control two states of expanded and contracted.

Figure 10: A reconfigurable spinning robot that folds its rods to make a more condensed shape.

Acknowledgements

I would like to thank Professor Zhao and the Zhao Lab for hosting me over the summer, my mentors Jeseung Lee, Lu Lu, and Sophie Leanza for their support and guidance, and the SURI program for the opportunity.

References:

[1] L. Lu, S. Leanza, J. Dai, J.W. Hutchinson, & R.R. Zhao, Multistability of segmented rings by programming natural curvature, Proc. Natl. Acad. Sci. U.S.A. 121 (31) e2405744121, <https://doi.org/10.1073/pnas.2405744121>, (2024).10.1073/pnas.2405744121.